Olympiad Test Paper — Mathematics (Class 7) — Paper 5

Chapter 14: Constructions and Tilings

Paper 5 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	14 — Constructions and Tilings
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Expert traps abound: 'close to 360°' or 'under 180°' never tiles, the bisection trick never trisects a general angle, the triangle inequality and a positive angle-sum must be checked before any triangle is drawn, perpendicular bisectors are equidistant from VERTICES while angle bisectors are equidistant from SIDES, an interior angle of a regular n-gon is 180−360/n and must stay strictly below 180°, and a leftover wedge that no regular polygon's interior angle can supply means the vertex simply cannot be completed. Do not write on this paper — mark answers on your answer sheet.

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1. Three radio masts A, B, C stand at the corners of a scalene triangle. A repeater is to be placed so that it is the same distance from all three masts AND inside the triangle. Which construction locates it, and for an obtuse triangle does the spot stay inside?

(A) Intersect the three angle bisectors (the incentre), which is equidistant from the three masts and always inside.
(B) Take the midpoint of the longest side, which is equidistant from all three vertices and always inside.
(C) Intersect the perpendicular bisectors of two sides (the circumcentre, equidistant from all three vertices); for an obtuse triangle that point lies OUTSIDE the triangle, so the 'inside' requirement cannot be met.
(D) Intersect the perpendicular bisectors; the circumcentre is always strictly inside every triangle, obtuse or not.

2. A chord AB of length 16 cm is drawn, and a circle through A and B has its centre O at distance 6 cm from AB. A second circle also passes through A and B but has radius 17 cm. The distance between the two possible centres of the second circle is:

(A) 30 cm, since each centre lies on the perpendicular bisector of AB at distance $\sqrt{(17^2-8^2)}=15$ cm from the chord, one on each side, giving 15+15.

(B) 15 cm, since $\sqrt{(17^2-8^2)}=15$ is the distance and there is only one centre.

(C) 9 cm, since 17−8=9.

(D) 34 cm, since 17+17=34.

3. From an external point F a perpendicular is dropped to line m, meeting it at N. Points P and Q lie on m with NP = NQ = 9 cm, and FN = 12 cm. The lengths FP and FQ are:

(A) Equal, each 21 cm, since FP = FN + NP = 12 + 9.

(B) Unequal, because P and Q lie on opposite sides of N.

(C) Equal, each $\sqrt{63}$ cm, since $FP^2 = 12^2 - 9^2 = 63$.

(D) Equal, each 15 cm, since FN ⊥ m makes FNP and FNQ congruent right triangles with $FP=\sqrt{(12^2+9^2)}=15$.

4. Point T lies on the perpendicular bisector of segment AB with TA = 25 cm; AB = 14 cm. A second point S also lies on that bisector with SA = 25 cm but on the opposite side of AB from T. The distance TS is:

(A) 24 cm, since $\sqrt{(25^2-7^2)}=24$ is the distance from the midpoint and T, S coincide there.

(B) 48 cm, since the bisector meets AB at its midpoint, the foot is $\sqrt{(25^2-7^2)}=24$ cm from each of T and S, and they sit on opposite sides: 24+24.

(C) 50 cm, since TA + SA = 25 + 25.

(D) 14 cm, the length of AB.

5. A student constructs the perpendicular bisectors of all three sides of a triangle and finds they meet at one point W. They then construct the three ANGLE bisectors. The two meeting points coincide only when the triangle is:

(A) Any triangle, since perpendicular bisectors and angle bisectors always meet at the same centre.

(B) Right-angled, since the right angle forces the two centres together.

(C) Isosceles, since two equal sides already align the two centres.

(D) Equilateral, because only then is the circumcentre (vertex-equidistant) the same point as the incentre (side-equidistant).

6. You must construct a 7.5° angle with compass and straightedge starting only from constructions in this chapter. The correct chain is:

(A) Trisect a 22.5° angle, since 22.5 ÷ 3 = 7.5.

(B) Construct 60° and trisect it twice, since 60 ÷ 9 ≈ 7.5.

(C) It cannot be constructed, because 7.5° is not a whole number of degrees.

(D) Construct 60°, bisect to 30°, bisect to 15°, bisect to 7.5° — three halvings of a constructed 60° angle.

7. Which angle below is the LARGEST that is impossible to construct with only a compass and straightedge?

(A) 75°, as 60° + 15°. (B) 52.5°, as half of the constructible 105°.

(C) 97.5°, as half of the constructible 195° (= 150° + 45°).

(D) 100°, since constructing it would mean trisecting a constructible 300° (or building 20°), and general trisection is impossible.

8. A right angle is constructed and then halved four times in succession, each time taking the new smaller angle. The smallest angle obtained and whether it is constructible are:

(A) 5.625°, but it is NOT constructible, since it is not a multiple of 15°.

(B) 7.5°, since $90 \div 12 = 7.5$.

(C) 5.625°, and it is constructible, since $90 \rightarrow 45 \rightarrow 22.5 \rightarrow 11.25 \rightarrow 5.625$ are four legitimate bisections.

(D) 11.25°, since only three halvings are possible before the angle is too small.

9. An angle of 84° is given. Using only bisection (never trisection), the SMALLEST angle reachable that is still a whole number of degrees is:

(A) 12°, since $84 \div 7 = 12$. (B) 28°, since $84 \div 3 = 28$ by trisection.

(C) 21°, since $84 \rightarrow 42 \rightarrow 21$, and the next halving (10.5°) is not a whole number.

(D) 10.5°, since $84 \rightarrow 42 \rightarrow 21 \rightarrow 10.5$ and that is the smallest.

10. To copy an angle, an arc of radius r cuts the arms at P and Q and the chord PQ is stepped off on a same-radius arc at the new vertex. A student instead uses a LARGER radius R at the new vertex but copies the SAME chord length PQ. The copied angle is:

(A) Exactly equal, since the chord PQ was copied faithfully.

(B) Larger than the original, since a larger radius opens the angle wider.

(C) Equal only if R is exactly double r.

(D) Smaller than the original, because the same chord across a larger-radius arc subtends a narrower angle.

11. A triangle has sides 11 cm and 6 cm. For how many INTEGER lengths (whole cm) of the third side does a genuine triangle exist?

(A) 13, since x runs from 5 to 17 inclusive.(B) 6, since $11 - 6 = 5$ gives the count.

(C) 11, since the third side x satisfies $5 < x < 17$, giving $x = 6,7,8,\ldots,16$.

(D) 9, since x runs from 7 to 15.

12. A triangle is to be constructed by ASA with base 7 cm and base angles 64° and 116°. Before drawing, feasibility is checked. The triangle:

(A) Cannot be built, since 64° + 116° = 180° leaves a 0° third angle, so the other two sides are parallel and never meet.

(B) Can be built; the third angle is 0°, which is allowed for a very flat triangle.

(C) Can be built, since the two base angles are supplementary and that guarantees a triangle.

(D) Cannot be built, because ASA needs the two base angles to be equal.

13. Using SAS you are given two sides 9 cm and 40 cm with the included angle 90°. The third side and the triangle's area are:

(A) Third side 49 cm and area 180 cm^2, since 9 + 40 = 49.

(B) Third side 41 cm and area 180 cm^2, since $\sqrt{(9^2+40^2)}=41$ and area = ½ × 9 × 40.

(C) Third side 41 cm and area 738 cm^2, since area = ½ × 9 × 41 ... using the hypotenuse.

(D) Third side $\sqrt{1519}$ cm, since $\sqrt{(40^2 - 9^2)} = \sqrt{1519}$.

14. Which single data set determines EXACTLY ONE triangle up to congruence?

(A) Two sides 8 cm and 5 cm with the 110° angle lying BETWEEN them.

(B) Two sides 8 cm and 5 cm with a 35° angle opposite the 8 cm side.

(C) Three angles 50°, 60° and 70°.

(D) Two sides 4 cm and 6 cm and the third side 10 cm.

15. A right triangle is built by RHS with hypotenuse 26 cm and one leg 10 cm. The triangle's PERIMETER and AREA are:

(A) Perimeter 52 cm and area 130 cm^2, since the other leg is 26 − 10 = 16.

(B) Perimeter 64 cm and area 180 cm^2, since the other leg is $\sqrt{(26^2+10^2)}$.

(C) It cannot be built, since 10 cm is shorter than half the hypotenuse.

(D) Perimeter 60 cm and area 120 cm^2, since the other leg is $\sqrt{(26^2-10^2)}=24$, perimeter 10+24+26, area ½×10×24.

16. A regular polygon has interior angle 174°. The number of sides is:

(A) 29, since 174 ÷ 6 = 29.

(B) 30, since 180 − 174 = 6 and 174 ÷ 6 ... gives near 30.

(C) 12, since 174° is close to a dodecagon's 150°.

(D) 60, since the exterior angle is 180 − 174 = 6° and 360 ÷ 6 = 60.

17. A regular polygon's interior angle is EXACTLY four times its exterior angle. The polygon is a:

(A) Regular octagon, since interior 135° is four times something.

(B) Regular decagon (10 sides), since interior = 4 × exterior with interior + exterior = 180 gives exterior 36°, so n = 360 ÷ 36 = 10.

(C) Regular square, since 'four times' points to four sides.

(D) Regular dodecagon, since exterior 30° and 30 × 4 = 120.

18. Regular polygon X has exterior angle 24° and regular polygon Y has interior angle 160°. The difference (number of sides of Y) − (number of sides of X) is:

(A) 0, since both are 15-sided.

(B) 3, since X has 360 ÷ 24 = 15 sides and Y has exterior 20° giving 360 ÷ 20 = 18 sides, and 18 − 15 = 3.

(C) 6, since Y has 21 sides and X has 15. (D) 33, since 15 + 18 = 33.

19. For which number of sides n is the interior angle of a regular n-gon EXACTLY three times its exterior angle?

(A) n = 6, since a hexagon's 120° is three times 40°.

(B) n = 3, since three times points to a triangle.

(C) n = 8, since interior = 3 × exterior with the two summing to 180 gives exterior 45°, and 360 ÷ 45 = 8.

(D) n = 12, since 12 = 3 × 4.

20. A regular polygon is such that 360° divided by its interior angle equals exactly 2.4. Its interior angle, number of sides, and tiling ability are:

(A) Interior 150°, a regular dodecagon, and it self-tiles with two corners (2.4 rounds to 2).

(B) Interior 150°, a regular dodecagon (12 sides), and it does NOT self-tile, since 2.4 is not a whole number.

(C) Interior 144°, a regular decagon, and it does not self-tile.

(D) Interior 150°, a regular pentadecagon (15 sides), and it does not self-tile.

21. Among the regular pentagon (108°), regular nonagon (140°), regular dodecagon (150°) and regular 18-gon (160°), exactly how many can self-tile the plane?

(A) None, since 360 ÷ 108, 360 ÷ 140, 360 ÷ 150 and 360 ÷ 160 are 3.33, 2.57, 2.4 and 2.25 — none whole.

(B) One, the dodecagon, since 150° is a multiple of 30°.

(C) Two, the dodecagon and the 18-gon, since both exceed 144°.

(D) All four, since every interior angle is below 180°.

22. The values of k (number of identical regular tiles meeting at a vertex) for which a regular polygon self-tiles are exactly 3, 4 and 6. A student asks why k = 5 is impossible. The reason is:

(A) k = 5 would need a regular pentagon, whose 108° angle gives 5 at a vertex.

(B) k = 5 is impossible because 5 is an odd number and tilings need even counts.

(C) k = 5 would need each interior angle to be 360 ÷ 5 = 72°, but no regular polygon has a 72° interior angle (the smallest is the triangle's 60°, the next is the square's 90°).

(D) k = 5 would need a 72° angle, supplied by half a regular pentagon.

23. At a tiling vertex the corner angles are measured and total 366°. The correct interpretation is:

(A) There is a 6° gap, since the total is near 360°.

(B) It is a valid vertex, since 366° is within rounding distance of 360°.

(C) The measurement is impossible, because vertex angles can never exceed 360°.

(D) The vertex does not close — there is a 6° OVERLAP, since the angles exceed 360°.

24. A semi-regular vertex is proposed: one regular dodecagon (150°), one regular hexagon (120°), and the rest filled with equilateral triangles (60°). Can it close, and with how many triangles?

(A) No — the leftover is 360 − 150 − 120 = 90°, and 90 ÷ 60 = 1.5 is not a whole number, so it does NOT close with triangles alone.

(B) Yes, with one triangle, since 360 − 150 − 120 = 60.

(C) Yes, with two triangles, since 2 × 60 = 120 fills the rest.

(D) Yes, with 1.5 triangles.

25. A vertex uses one regular octagon (135°), one square (90°), and the rest filled with equilateral triangles (60°) only. How many triangles close it exactly?

(A) Two triangles, since 360 − 135 − 90 = 120 and 120 ÷ 60 = 2.

(B) It cannot be closed with triangles, since 360 − 135 − 90 = 135 and 135 ÷ 60 = 2.25 is not a whole number.

(C) Three triangles, since 3 × 60 = 180 fills the rest.

(D) 2.25 triangles, which is allowed in semi-regular tilings.

26. Three DIFFERENT regular polygons meet at a vertex: a square (90°), a regular hexagon (120°), and a third polygon that closes the vertex exactly. The third polygon is a:

(A) Regular hexagon (120°), since 360 − 90 − 120 = 150 rounds to 120.

(B) Regular octagon (135°), since the leftover is about 135°.

(C) Regular polygon of interior angle 150°, but no such polygon exists.

(D) Regular dodecagon (150°), since 360 − 90 − 120 = 150°, whose exterior 30° gives 12 sides.

27. A vertex holds one regular dodecagon (150°) and the rest is filled by k equal regular polygons. Checking k = 1, 2, 3, the verdict is:

(A) k = 3 hexagons, since 360 − 150 = 210 and 210 ÷ 3 = 70 ≈ 120.

(B) It cannot be completed, since the 210° leftover would need each tile to be 210° (k=1), 105° (k=2) or 70° (k=3) — none of which is a real regular interior angle.

(C) k = 2 squares, since 210 ÷ 2 = 105 ≈ 90.

(D) k = 1 polygon of 210°, a valid regular polygon.

28. At a vertex one regular pentagon (108°) is placed. A student wants to fill the rest with squares (90°) only. The verdict is:

(A) Impossible, since 360 − 108 = 252 and 252 ÷ 90 = 2.8 is not a whole number.

(B) Three squares, since 3 × 90 = 270 ≈ 252.

(C) Two squares, since 2 × 90 = 180 and the rest is ignored.

(D) 2.8 squares, which is allowed at a semi-regular vertex.

29. A quadrilateral tile has angles 70°, 95°, 100° and 95°. Asked whether copies of THIS quadrilateral tile the plane, the correct fact is:

(A) Yes — every quadrilateral tiles, because its four angles sum to 360°, so arranging one of each angle around a point closes the vertex; rotating copies fills the plane.

(B) No — only regular polygons tile, and this one is irregular.

(C) No — its angles must each individually divide 360° for it to tile.

(D) Only if it is a parallelogram.

30. A vertex is to be closed using a triangle (60°), a square (90°), and THREE more equal regular polygons. Each of those three must have interior angle:

(A) 70°, supplied by a regular polygon between a triangle and a square.

(B) 70°, which no regular polygon supplies (the triangle's 60° is the smallest), so the configuration cannot be completed with three equal regular tiles.

(C) 60°, three equilateral triangles. (D) 90°, three squares.

31. At a vertex one square (90°), one regular hexagon (120°), and the rest filled with equilateral triangles (60°). How many triangles close it exactly?

(A) Two triangles, since $360 - 90 - 120 = 150$ and $150 \div 60 \approx 2$.

(B) Three triangles, since $3 \times 60 = 180$ fills the rest.

(C) It cannot be closed with triangles, since $360 - 90 - 120 = 150$ and $150 \div 60 = 2.5$ is not a whole number.

(D) 2.5 triangles, which is permitted.

32. Six equilateral triangles tile around a point. If you remove TWO adjacent triangles and try to fill the resulting gap with a single regular polygon, that polygon would need interior angle:

(A) 60°, just another triangle.

(B) 240°, which is impossible for a regular polygon.

(C) 180°, which is impossible for a regular polygon.

(D) 120°, a regular hexagon, since removing two 60° triangles opens a 120° gap.

33. A perpendicular bisector of segment CD is drawn, and at C a 90° angle to CD is also constructed. A student claims the perpendicular-at-C and the perpendicular bisector of CD are the same line. The truth is:

(A) They are the same line, since both are perpendicular to CD.

(B) They cross at the midpoint of CD.

(C) They are different PARALLEL lines, both perpendicular to CD but meeting CD at different points (C versus the midpoint).

(D) They cross at C.

34. You construct a 60° angle, a 90° angle, and a 45° angle (half of 90°), each sharing the same vertex and arm in turn so they stack. The largest single angle you can make by adding constructible pieces that is still BELOW 180° and a whole number of degrees, using 60° and 45° pieces only, is:

(A) 165°, as 60° + 60° + 45°. (B) 150°, as 60° + 45° + 45°.

(C) 180°, as 60° + 60° + 60°. (D) 170°, as 45° + 45° + 80°.

35. A triangle ABC is built by SAS with AB = AC = 7 cm and included angle A = 120°. Side BC, compared with AB, is:

(A) Longer than AB, specifically BC = $7\sqrt{3}$ cm, since a wide 120° apex spreads B and C far apart.

(B) Equal to AB, making the triangle equilateral.

(C) Shorter than AB, since equal sides force a short base.

(D) Exactly 14 cm, since 7 + 7 = 14.

36. A triangle has two sides 5 cm and 13 cm and the angle between them is steadily increased from 20° toward 160°. The third side (opposite that angle):

(A) Stays constant, since the two given sides are unchanged.

(B) Decreases throughout, since a wider angle folds the ends closer.

(C) Increases throughout, growing from near 8 cm toward near 18 cm as the included angle widens.

(D) First decreases then increases, with a minimum at 90°.

37. A triangle has sides 7 cm and 24 cm with a 90° angle OPPOSITE the longest side. The third (longest) side and the triangle's area are:

(A) Longest side 31 cm and area 84 cm^2, since 7 + 24 = 31.

(B) Hypotenuse 25 cm and area 300 cm^2, since area = ½ × 24 × 25.

(C) Hypotenuse $\sqrt{527}$ cm, since $\sqrt{(24^2 - 7^2)} = \sqrt{527}$.

(D) Hypotenuse 25 cm and area 84 cm^2, since $\sqrt{(7^2+24^2)}=25$ and area = ½ × 7 × 24.

38. Which statement about constructibility and trisection is COMPLETELY true?

(A) Any angle can be trisected by swinging three equal arcs along an arm.

(B) No angle can ever be divided into three equal parts with these tools.

(C) Trisection works only for angles above 90°.

(D) A general angle cannot be trisected with compass and straightedge, yet a 90° angle CAN be split into three 30° parts (because 30° is independently constructible).

39. A regular polygon's interior angle exceeds its exterior angle by 140°. The polygon is a:

(A) Regular dodecagon, since the difference 140 points near 150°.

(B) Regular 18-gon, since interior − exterior = 140 and interior + exterior = 180 give exterior 20°, so 360 ÷ 20 = 18.

(C) Regular decagon, since exterior 20° gives 10 sides.

(D) Regular nonagon, since 140° is a nonagon's interior angle.

40. At a vertex one regular dodecagon (150°), one square (90°), and the rest filled with equilateral triangles (60°). How many triangles close it exactly?

(A) One triangle, since 360 − 150 − 90 = 60.

(B) Three triangles, since 3 × 60 = 180 fills the rest.

(C) It cannot close, since the leftover is not a multiple of 60.

(D) Two triangles, since 360 − 150 − 90 = 120 and 120 ÷ 60 = 2.

41. A semi-regular vertex uses two regular hexagons (120° each) and the rest filled with equilateral triangles (60°). How many triangles close it, and is the configuration valid?

(A) One triangle, since 360 − 240 = 60.

(B) Two triangles, valid, since 360 − 240 = 120 and 120 ÷ 60 = 2.

(C) Three triangles, since 3 × 60 = 180 fills the rest.

(D) It cannot close, since two hexagons already use 240°.

42. A regular polygon has interior angle A. The same polygon is used with squares so that at a vertex exactly one such polygon and two squares (90° each) meet and close it. The polygon is a:

(A) Regular polygon of interior 180°, which is impossible — so one polygon plus two squares can NEVER close a vertex with a real regular tile.

(B) Regular hexagon (120°), since 360 − 180 = 180 ≈ 120.

(C) Regular dodecagon (150°), since 360 − 180 = 180 ≈ 150.

(D) Square (90°), making four squares total.

43. A construction is called EXACT (not merely a neat drawing) because:

(A) The compass measures lengths to the nearest millimetre.

(B) The pencil lines are drawn infinitely thin.

(C) Its correctness rests on geometric facts about equal radii and equidistance, so it is right in principle however unsteady the hand — not on careful measuring.

(D) The figure is traced over a printed template.

44. Three regular polygons meet at a vertex: a square (90°), a regular octagon (135°), and one more polygon that closes the vertex. The third polygon is a:

(A) Regular octagon (135°), since 360 − 90 − 135 = 135° — the classic 4.8.8 vertex.

(B) Regular hexagon (120°), since 360 − 90 − 135 = 135 ≈ 120.

(C) Square (90°), since 360 − 90 − 135 = 135 ≈ 90.

(D) Regular polygon of interior 135°, but no such polygon exists.

45. A regular polygon has interior angle 162°. If you try to tile with it alone, the gap or overlap left by the nearest whole numbers of tiles is:

(A) Two tiles give exactly 360°, so it tiles.

(B) Two tiles give 324° (a 36° gap) and three give 486° (a 126° overlap); since 360 ÷ 162 ≈ 2.22 is not whole, it cannot self-tile.

(C) Three tiles give exactly 360°, so it tiles.(D) Two tiles give a 6° gap, so it nearly tiles.

46. Equilateral triangles tile the plane. A student claims that therefore EVERY isosceles triangle tiles, but no scalene triangle does. The correct fact is:

(A) Only isosceles triangles tile; scalene ones leave gaps.

(B) Only triangles whose angles each divide 360° tile.

(C) Every triangle of any shape tiles, since two copies of any triangle join along a side into a parallelogram, and parallelograms tile by sliding.

(D) Only acute triangles tile; obtuse ones overlap.

47. A triangle is to be constructed given two sides 6 cm and 6 cm and a 50° angle OPPOSITE one of the 6 cm sides. The number of non-congruent triangles fitting this data is:

(A) Two, since SSA is always the ambiguous case.

(B) None, since SSA never determines a triangle.

(C) Exactly one, since the two sides are equal so the SSA ambiguity collapses to a single isosceles triangle.

(D) Infinitely many, since two angles are equal.

48. A vertex uses one square (90°) and the rest filled by equilateral triangles (60°) only. The only whole number of triangles that closes it is:

(A) Four triangles, since $4 \times 60 = 240$ and $240 + 90 = 330 \approx 360$.

(B) It cannot be closed by triangles alone, since $360 - 90 = 270$ and $270 \div 60 = 4.5$ is not a whole number.

(C) Five triangles, since $5 \times 60 = 300$ and $300 + 90 = 390 \approx 360$.

(D) 4.5 triangles, which is permitted.

49. A regular polygon's exterior angle, added to the exterior angle of an equilateral triangle (120°), gives 165°. That polygon is a:

(A) Regular hexagon, since exterior 60° gives 6 sides.

(B) Regular octagon, since exterior 45° gives $360 \div 45 = 8$ sides.

(C) Regular pentagon, since 45° is close to a pentagon's 72°.

(D) Regular dodecagon, since exterior 30° gives 12 sides.

50. Two regular polygons of the SAME kind plus one square (90°) meet at a vertex and close it exactly. Each of the two equal polygons has interior angle:

(A) 120°, a regular hexagon, since $(360 - 90) \div 2 = 135 \approx 120$.

(B) 150°, a regular dodecagon, since two of them plus a square close the vertex.

(C) 135°, a regular octagon, since $(360 - 90) \div 2 = 135$ — the 4.8.8 vertex.

(D) 135°, but no regular polygon has that interior angle.

51. A right triangle is constructed by RHS with hypotenuse 15 cm and one leg 9 cm. A second is built with hypotenuse 15 cm and leg 12 cm. The difference in their areas is:

(A) 0 cm^2, since the first has legs 9 and 12 (area 54) and the second has legs 12 and 9 (area 54) — they are congruent.

(B) 54 cm^2, since the first area is 54 and the second is 0.

(C) 27 cm^2, since the areas are 54 and 27.

(D) It cannot be built, since 12 cm exceeds the hypotenuse 15 cm minus 9 cm.

52. A semi-regular vertex repeats: triangle (60°), square (90°), triangle (60°), square (90°), triangle (60°). Does this close to 360°?

(A) No, it leaves a 30° gap, since the total is 330°.

(B) No, it overlaps by 60°, since the total is 420°.

(C) Yes — the angles total $3 \times 60 + 2 \times 90 = 180 + 180 = 360°$, so it closes exactly.

(D) It cannot be a vertex, since five tiles never fit.

53. A regular polygon has interior angle A with $120° < A < 150°$. Can such a polygon EVER self-tile the plane?

(A) No, because $360 \div A$ then lies strictly between 2.4 and 3, so it can never be a whole number.

(B) Yes, if A is a multiple of 15°, such as 135°.

(C) Yes, for any A in that range, since two or three corners can fit.

(D) No, because A in that range exceeds 180°.

54. To construct a 52.5° angle with compass and straightedge, a valid route is:

(A) Trisect a 157.5° angle to get 52.5° directly.

(B) Construct 105° as 60° + 45° (45° being half of a 90°), then bisect 105° to get 52.5°.

(C) Bisect a 60° angle, since $60 \div 2 \approx 52.5$.

(D) It cannot be constructed, since 52.5° is not a multiple of 15°.

55. A perpendicular bisector of AB meets AB at M. A point P is taken on the bisector with PA = 10 cm, and a point Q on AB with AQ = 6 cm; AB = 16 cm. Comparing PA and PQ:

(A) PA = PQ, since P is equidistant from all points.

(B) PA = 10 cm and $PQ = \sqrt{(PM^2 + MQ^2)}$ where $PM = \sqrt{(10^2-8^2)} = 6$ and $MQ = |8-6| = 2$, giving $PQ = \sqrt{40} = 2\sqrt{10} \approx 6.3$ cm, so PQ < PA.

(C) PQ = 4 cm, since AQ = 6 and AM = 8 give MQ = 2 ... then PQ = 2.

(D) PQ = 10 cm, the same as PA.

56. A regular polygon and a regular hexagon (120°) and a square (90°) are arranged with TWO copies of the regular polygon plus the hexagon plus the square at one vertex, closing it. Each copy of the polygon has interior angle:

(A) 75°, which no regular polygon supplies, so this exact vertex cannot be completed with two equal regular tiles.

(B) 75°, supplied by a regular polygon between a triangle and a square.

(C) 60°, two equilateral triangles. (D) 90°, two more squares.

57. Among regular polygons, exactly three self-tile. A student lists, for each, the value of $360 \div$ (interior angle). The correct list of those three quotients is:

(A) 3, 4 and 5 — for triangle, square and pentagon.

(B) 2, 3 and 4 — for hexagon, ... and squares.

(C) 6, 4 and 3.33 — for triangle, square and pentagon.

(D) 6, 4 and 3 — for the triangle ($360 \div 60$), square ($360 \div 90$) and hexagon ($360 \div 120$).

58. A triangle is constructed by SAS with sides 6 cm and 6 cm and included angle 60°. The triangle is:

(A) Isosceles but not equilateral, with base shorter than 6 cm.

(B) Right-angled, with hypotenuse $6\sqrt{2}$ cm.

(C) Equilateral with all sides 6 cm, since two equal sides and a 60° included angle force the third side to also be 6 cm.

(D) Impossible, since equal sides cannot bound a 60° angle.

59. A vertex is closed by one regular 18-gon (160°), one regular polygon X, and one equilateral triangle (60°). For the vertex to close exactly, X's interior angle is:

(A) 150°, a regular dodecagon, since $360 - 160 - 60 = 140 \approx 150$.

(B) 120°, a regular hexagon, since $360 - 160 - 60 = 140 \approx 120$.

(C) 140°, but no regular polygon has that interior angle.

(D) 140°, a regular nonagon (9 sides), since $360 - 160 - 60 = 140$ and exterior 40° gives $360 \div 40 = 9$.

60. The single rule that decides whether ANY collection of tiles (regular, semi-regular, triangles, or general quadrilaterals) fits around every interior vertex is that the corner angles meeting there sum to:

(A) Exactly 360° — the full turn; any less leaves a gap and any more forces an overlap.

(B) Exactly 180°, a straight angle.

(C) At most 360°, so small gaps are tolerated.

(D) Any multiple of 90°, such as 270° or 450°.

Solutions — Mathematics (Class 7), Chapter 14: Constructions and Tilings — Paper 5

Paper 5 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	C	21	A	31	C	41	B	51	A
2	A	12	A	22	C	32	D	42	A	52	C
3	D	13	B	23	D	33	C	43	C	53	A
4	B	14	A	24	A	34	A	44	A	54	B
5	D	15	D	25	B	35	A	45	B	55	B
6	D	16	D	26	D	36	C	46	C	56	A
7	D	17	B	27	B	37	D	47	C	57	D
8	C	18	B	28	A	38	D	48	B	58	C
9	C	19	C	29	A	39	B	49	B	59	D
10	D	20	B	30	B	40	D	50	C	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) The point equally far from three vertices is the circumcentre, found where perpendicular bisectors of the sides cross. The incentre (angle bisectors) is equidistant from the SIDES, not the vertices, so that choice mixes up the two loci. The midpoint of one side is equidistant only from that side's two endpoints. The circumcentre lies outside an obtuse triangle, so the 'inside' condition fails — the catch in this item.

2. (A) Any centre of a circle through A and B sits on the perpendicular bisector of AB. The half-chord is 8 cm, so the centre's distance from the chord is $\sqrt{(17^2-8^2)}=\sqrt{225}=15$ cm. There are two such points, one on each side of AB, separated by 15+15=30 cm. The 6 cm detail about O is a distractor that plays no role. 15 forgets the second centre; 9 and 34 are raw add/subtract traps.

3. (D) FN is perpendicular to m, so triangles FNP and FNQ are right-angled at N with equal legs (FN common, NP=NQ). Hence $FP=FQ=\sqrt{(12^2+9^2)}=\sqrt{225}=15$ cm. Adding lengths along a bent path (12+9) is wrong because FP is the hypotenuse, not a straight sum. Symmetry forces equality regardless of side. Subtracting the squares confuses this with a different right-triangle setup.

4. (B) The perpendicular bisector crosses AB at the midpoint M, with AM=7. In right triangle TMA, TM=$\sqrt{(25^2-7^2)}=\sqrt{576}$=24 cm; likewise SM=24. As S and T are on opposite sides, TS=TM+SM=48 cm. 24 forgets the second point; 50 wrongly adds the slant lengths TA, SA; 14 is the base, unrelated.

5. (D) Perpendicular bisectors meet at the circumcentre; angle bisectors meet at the incentre. These are generally different points (one equidistant from vertices, the other from sides). They coincide only in the fully symmetric case — the equilateral triangle. Isosceles symmetry puts both on one axis but not at the same point, and a right angle does nothing special here.

6. (D) Repeated bisection is always allowed: 60 → 30 → 15 → 7.5. Trisection of a general angle (including 22.5°) is impossible with these tools, so the first wrong route fails; the second invents a 'trisect twice' that is equally impossible. Non-integer measures are perfectly constructible (15° already is half of 30°), so the last option is false.

7. (D) 75° = 60° + (half of 30°) is constructible; 52.5° halves 105° = 60°+45°; 97.5° halves 195° = 150°+45° — all built from constructible parts by addition and bisection. 100° = 5 × 20°, and 20° is the classic non-constructible angle (it would trisect 60°), so 100° cannot be made. It is also the largest of the listed values, matching 'largest impossible'.

8. (C) Each bisection halves the angle: 90, 45, 22.5, 11.25, 5.625. Four halvings reach 5.625°, and bisection of any constructible angle is always allowed, so it IS constructible — being a multiple of 15° is not required. 90÷12 is an invented relation; there is no limit forcing only three halvings.

9. (C) Bisecting 84° gives 42°, then 21°; halving again gives 10.5°, which is not a whole number. So the smallest WHOLE-number angle reachable by bisection alone is 21°. 12° would need a divide-by-7 that bisection cannot do; 28° requires trisection, which is impossible; 10.5° is reachable but is not a whole number, so it fails the stated condition.

10. (D) The angle depends on the chord-to-radius ratio, not the chord alone. A fixed chord on a bigger circle subtends a smaller central angle, so the copy comes out smaller. Faithful copying needs the SAME radius. Bigger radius does not 'open' the angle; the equal-only-if-double claim is invented.

11. (C) By the triangle inequality $11-6 < x < 11+6$, i.e. $5 < x < 17$. Whole numbers strictly between 5 and 17 are 6 through 16, which is 16−6+1 = 11 values. Counting 5 and 17 inclusive (13) wrongly allows degenerate triangles; the other two are arbitrary.

12. (A) The two base angles already use up all 180°, leaving 0° for the apex, so the arms drawn from the ends of the base are parallel and never meet — no triangle. A 0° angle is not a genuine triangle. Supplementary base angles are exactly the failing case, not a guarantee, and ASA does not require equal angles.

13. (B) With the right angle between the two given sides, they are the legs: third side = $\sqrt{(9^2+40^2)}=\sqrt{1681}=41$ cm, and area = ½ × leg × leg = ½ × 9 × 40 = 180 cm^2. Adding the legs (49) is wrong for a hypotenuse; using the hypotenuse as a height inflates the area; subtracting the squares is the wrong Pythagorean direction.

14. (A) Two sides with the included angle (SAS) fix a unique triangle. An angle opposite a side is the ambiguous SSA case (can give two triangles or none). Three angles (AAA) fix only shape, not size — infinitely many. Sides 4, 6, 10 fail the triangle inequality (4+6 = 10), so no triangle exists at all.

15. (D) Other leg = $\sqrt{(26^2-10^2)}=\sqrt{576}=24$ cm. Perimeter = 10+24+26 = 60 cm; area = ½×10×24 = 120 cm^2. Subtracting raw lengths (26−10) is wrong; adding squares uses the wrong Pythagorean direction; and there is no rule forbidding a leg shorter than half the hypotenuse — RHS only needs the leg shorter than the hypotenuse.

16. (D) Exterior angle = 180 − interior = 6°, and the number of sides = 360 ÷ exterior = 360 ÷ 6 = 60. The other options divide the wrong quantities or guess by closeness to a known angle, none of which gives the sides.

17. (B) Let exterior = e; interior = 4e and interior + exterior = 180 give 5e = 180, e = 36°. Then n = 360 ÷ 36 = 10, a decagon. The octagon's exterior is 45°, not a 1:4 ratio; 'four' does not mean four sides; for the dodecagon the ratio interior:exterior is 150:30 = 5:1, not 4:1.

18. (B) X: sides = 360 ÷ 24 = 15. Y: exterior = 180 − 160 = 20°, so sides = 360 ÷ 20 = 18. Difference 18 − 15 = 3. The other options miscompute Y's side count or add instead of subtract.

19. (C) interior = 3·exterior and interior + exterior = 180 give 4·exterior = 180, exterior = 45°, so n = 360 ÷ 45 = 8 (octagon). A hexagon's exterior is 60°, and 120 = 2 × 60 (ratio 2:1, not 3:1). The other options guess from the number three with no calculation.

20. (B) 360 ÷ angle = 2.4 means angle = 360 ÷ 2.4 = 150°, whose exterior is 30°, so n = 360 ÷ 30 = 12 (dodecagon). Since 2.4 is not whole, it cannot self-tile; rounding the divisor never produces a real tiling. 144° corresponds to 2.5, and a 15-gon's interior is 156°, not 150°.

21. (A) Self-tiling needs 360 ÷ interior to be a whole number. Here the quotients are 3.33, 2.57, 2.4 and 2.25 — none is an integer, so none tiles. Being a multiple of 30° or under 180° is irrelevant; only the only-triangle-square-hexagon trio (60°, 90°, 120°) divides 360° evenly.

22. (C) k tiles of interior angle A close a vertex when kA = 360, so A = 360/k. For k = 5, A = 72°, but regular interior angles jump 60° (triangle), 90° (square), 108° (pentagon)... — none is 72°, so no regular polygon fits. A pentagon's 108° gives 360÷108 = 3.33, not 5; oddness is irrelevant; and tiles must be whole regular polygons, not halves.

23. (D) Angles meeting at an interior vertex must sum to exactly 360°. More than 360° means tiles pile on top of one another — an overlap of 6° here. Less than 360° would be a gap; 366° is more, so it overlaps. Rounding is never allowed, and an attempted arrangement certainly can total more than 360° (that is precisely what overlap means).

24. (A) The leftover after the dodecagon and hexagon is 360 − 150 − 120 = 90°. Filling 90° with 60° triangles needs 90 ÷ 60 = 1.5 — not a whole number — so the vertex cannot be completed with triangles alone (a single square's 90° would close it, but not triangles). The wrong options misadd the leftover or accept a fractional tile count.

25. (B) Leftover = 360 − 135 − 90 = 135°. Dividing by 60° gives 2.25, not a whole number, so triangles cannot complete the vertex. The first wrong option misadds (135 + 90 = 225, leftover 135, not 120); the second overshoots; and fractional tiles are never permitted.

26. (D) Leftover = 360 − 90 − 120 = 150°. A regular polygon with interior 150° has exterior 30° and 360 ÷ 30 = 12 sides — a dodecagon, which exists. Rounding 150 to 120 or guessing 135 is wrong, and the claim that no 150° polygon exists is false (the dodecagon does).

27. (B) Leftover = 360 − 150 = 210°. To fill with k equal tiles, each interior angle is 210/k: k=1 gives 210° (over 180°, impossible), k=2 gives 105° (no regular polygon), k=3 gives 70° (no regular polygon). None is a real regular interior angle, so the vertex cannot be completed by equal regular tiles. The distractors round impossible angles to nearby real ones.

28. (A) Leftover = 360 − 108 = 252°. Dividing by a square's 90° gives 2.8, not whole, so squares cannot complete the vertex. Three squares would overshoot (270 > 252) and two would leave a gap; fractional tiles are never permitted.

29. (A) Any quadrilateral tiles the plane: its angle sum is 360°, so placing one copy of each of the four angles around a vertex exactly closes it, and repeating fills the plane. Tiling does not require regularity, nor that each angle divide 360°, nor that the shape be a parallelogram.

30. (B) Leftover = 360 − 60 − 90 = 210°; split into three equal tiles gives 210 ÷ 3 = 70° each. No regular polygon has a 70° interior angle (the smallest possible is 60°), so this cannot be done. Three triangles would give 180° (leftover wrong), three squares 270° (overshoot).

31. (C) Leftover = 360 − 90 − 120 = 150°. Dividing by 60° gives 2.5 — not whole — so triangles cannot complete it (two give a 30° gap, three overshoot by 30°). Fractional tiles are never allowed.

32. (D) Two adjacent triangles cover 2 × 60 = 120°. Removing them leaves a 120° wedge, which exactly fits a regular hexagon's interior angle. A single triangle would leave 60° (only fills one), and 240° or 180° exceed the gap and are impossible regular angles anyway.

33. (C) Two distinct lines both perpendicular to the same segment CD are parallel to each other. The perpendicular bisector meets CD at its midpoint; the perpendicular at C meets it at C. Being perpendicular to CD does not make them the same line, and parallel lines do not cross (so neither 'cross' option holds).

34. (A) Adding copies of 60° and 45°: 60+60+45 = 165°, which is below 180° and a whole number. 60+45+45 = 150° is smaller; 60×3 = 180° is not below 180°; and 80° is not a 60°-or-45° piece, so 170° is not built from the allowed pieces.

35. (A) With apex 120° and equal arms, the base BC is the longest side. (By the cosine relation $BC^2 = 7^2 + 7^2 - 2\cdot7\cdot7\cdot\cos120° = 49+49+49 = 147$, so $BC = 7\sqrt{3} \approx 12.1$ cm > 7.) A 120° apex is not equilateral (that needs 60°); equal arms do not force a short base; and 14 cm would require the points to be collinear (180°), not 120°.

36. (C) Widening the included angle pushes the two far endpoints apart, so the opposite side grows monotonically (from near $|13-5| = 8$ toward near 13+5 = 18). It cannot stay constant, does not shrink, and has no minimum at 90° — the relationship is strictly increasing across 0° to 180°.

37. (D) The right angle is opposite the longest side, so that side is the hypotenuse $\sqrt{(7^2+24^2)}=\sqrt{625}=25$ cm; 7 and 24 are the perpendicular legs giving area = ½ × 7 × 24 = 84 cm^2. Adding the legs (31) is not a hypotenuse; using the hypotenuse as a height inflates the area; subtracting squares is the wrong direction.

38. (D) General trisection is impossible, but special angles can still be split into thirds when each third is itself constructible — 90° into three 30°s works because 30° is constructible (half of 60°). Equal arcs along an arm divide the arm, not the angle, so the first claim is the classic fallacy; trisection is sometimes possible (the special cases), refuting 'no angle ever'; and there is no 'above 90°' rule.

39. (B) From interior − exterior = 140 and interior + exterior = 180, adding gives 2·interior = 320, interior = 160°, exterior = 20°. Sides = 360 ÷ 20 = 18. The decagon's exterior is 36° (not 20°); the nonagon's interior is 140° but its exterior is 40°, so the difference would be 100°, not 140°.

40. (D) Leftover = 360 − 150 − 90 = 120°, and 120 ÷ 60 = 2, so exactly two triangles close the vertex (this is the real 3.4.3.12-type family). The first wrong option misadds the leftover; three overshoot; and 120 IS a multiple of 60, so the 'cannot close' claim is false.

41. (B) Two hexagons cover 240°, leaving 360 – 240 = 120°, which two 60° triangles fill exactly (the 3.3.6.6 / 3.6.3.6 family). One triangle leaves a gap, three overshoot, and 240° does leave room for exactly 120° more — so it does close.

42. (A) Two squares contribute 180°, leaving 360 – 180 = 180° for the single extra polygon. No regular polygon has a 180° interior angle (that would be a straight line), so this vertex cannot be completed by one more regular tile. Rounding 180 down to 120 or 150 is invalid, and 'four squares' changes the stated count of two.

43. (C) A construction is exact because the steps guarantee the result by geometry (equal radii give equal distances, equidistant points lie on bisectors), independent of measurement precision. It does not rely on millimetre measuring, infinitely thin lines, or tracing a template.

44. (A) Leftover = 360 – 90 – 135 = 135°, which is exactly another regular octagon's interior angle — the well-known square-octagon-octagon (4.8.8) tiling. The octagon does exist (exterior 45°, 360 ÷ 45 = 8), so the 'no such polygon' claim is false; rounding to 120 or 90 is wrong.

45. (B) 360 ÷ 162 ≈ 2.22, not whole. Two tiles: 2 × 162 = 324°, a 36° gap; three tiles: 486°, a 126° overlap. Neither closes, so it cannot self-tile. The exact-360 claims are false, and the 6° gap is a miscalculation.

46. (C) Any triangle tiles: rotate a copy 180° about the midpoint of one side to form a parallelogram, and parallelograms tile by translation. So scalene, obtuse, and isosceles all tile equally — the shape restriction in the wrong options is false, as is the 'angles must divide 360°' rule (that applies only to a single regular polygon meeting copies of itself).

47. (C) SSA is usually ambiguous, but with the two sides equal the angle opposite the second equal side must equal the given 50°, fixing a unique isosceles triangle — the ambiguity collapses. So exactly one triangle exists. SSA is not 'always' two or 'never' valid, and nothing here gives infinitely many.

48. (B) Leftover after one square = 360 – 90 = 270°; dividing by 60° gives 4.5 — not whole — so triangles alone cannot close it (four leave a 30° gap, five overshoot by 30°). Fractional tiles are never allowed. The 4.4.3.3.3 valid vertex needs TWO squares (180°) plus three triangles, not one square.

49. (B) If the two exterior angles add to 165° and the triangle's is 120°, the other is 165 – 120 = 45°, giving 360 ÷ 45 = 8 sides — an octagon. A hexagon's exterior is 60°, a pentagon's 72°, a dodecagon's 30° — none is 45°.

50. (C) Leftover after the square = 360 − 90 = 270°, split into two equal tiles gives 135° each — regular octagons (exterior 45°, eight sides). This is the 4.8.8 tiling. The octagon does exist, so the 'no such polygon' claim is false; 120° and 150° do not satisfy (360−90)/2 = 135.

51. (A) First triangle: other leg = $\sqrt{15^2-9^2}=12$, area = $\frac{1}{2}\times9\times12 = 54$. Second: other leg = $\sqrt{15^2-12^2}=9$, area = $\frac{1}{2}\times12\times9 = 54$. They are the same 9-12-15 triangle, so the area difference is 0. 12 cm is less than the 15 cm hypotenuse, so it is perfectly constructible — no rule about 15 − 9 applies.

52. (C) Three triangles give 180° and two squares give 180°, totalling 360° exactly — the vertex closes (this is a genuine semi-regular pattern). The wrong options misadd to 330 or 420, and there is no rule against five tiles meeting at a point (six triangles already do).

53. (A) For 120° < A < 150°, 360 ÷ A falls between 360÷150 = 2.4 and 360÷120 = 3, so it is strictly between 2.4 and 3 and never a whole number — no self-tiling. Being a multiple of 15° (e.g. 135°, giving 2.67) does not help, and A in this range is well below 180°, so that reason is wrong.

54. (B) 105° = 60° + 45° is constructible, and bisecting any constructible angle is always allowed, giving 105 ÷ 2 = 52.5°. Trisection of a general angle is impossible, half of 60° is 30° (not 52.5°), and being a multiple of 15° is not required for constructibility.

55. (B) M is the midpoint, AM = 8. In right triangle PMA, $PM = \sqrt{10^2-8^2} = 6$. Q lies 6 cm from A on AB, so MQ = 8 − 6 = 2. Then $PQ = \sqrt{PM^2+MQ^2} = \sqrt{36+4} = \sqrt{40} = 2\sqrt{10} \approx$ 6.3 cm, less than PA = 10. P is equidistant only from A and B (the segment endpoints), not from every point, so PA = PQ is false; PQ is not simply MQ.

56. (A) Leftover after hexagon and square = 360 − 120 − 90 = 150°; split into two equal tiles gives 75° each. No regular polygon has a 75° interior angle (the smallest is 60°), so the vertex cannot be completed this way. Two triangles give 120° (not 150°) and two squares 180° (overshoot).

57. (D) The three self-tilers have interior angles 60°, 90°, 120°, giving 360÷60 = 6, 360÷90 = 4, 360÷120 = 3 — all whole. The pentagon's 360÷108 = 3.33 is not whole (so it does not tile), ruling out lists that include it; '2,3,4' misassigns the hexagon's quotient.

58. (C) With two sides 6 cm and the included angle 60°, the base $BC^2 = 6^2+6^2-2\cdot6\cdot6\cdot\cos60° = 36+36-36 = 36$, so BC = 6 — all sides equal, an equilateral triangle. A 60° apex with equal arms is precisely equilateral, not merely isosceles, and not right-angled; the construction is certainly possible.

59. (D) Leftover = 360 − 160 − 60 = 140°. A regular polygon with interior 140° has exterior 40° and 360 ÷ 40 = 9 sides — a nonagon, which exists. Rounding 140 to 150 or

120 is wrong, and the nonagon does have a 140° interior angle, so the 'no such polygon' claim is false.

60. (A) Going once around an interior vertex is a full turn of 360°, so the corner angles meeting there must total exactly 360°. Less than that leaves an uncovered wedge (gap); more piles tiles on top of one another (overlap). 180°, 'at most 360°', and 'any multiple of 90°' all violate this exact-turn requirement.